

PMLZA 1. Intro and Overview

UoP

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ICMAT-CSIC

Overview

Overview

- **Machine learning (ML)** studies algorithms that automatically improve through experience in performing a task. These algorithms build statistical models from sample data (the **training data**) to make predictions and support decision-making.

Widely used in numerous real-world applications today.

- **Probabilistic machine learning (PML)** is an approach to machine learning that represents uncertainty explicitly using probability theory

Machine learning

From Wikipedia

ML: the study of computer algos that improve automatically through experience. It is seen as a part of AI. ML algos build a model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to do so....

A subset of ML is closely related to computational statistics, which focuses on making predictions using computers; but not all ML is statistical learning.

The study of mathematical optimization delivers methods, theory and application domains to the field of machine learning.

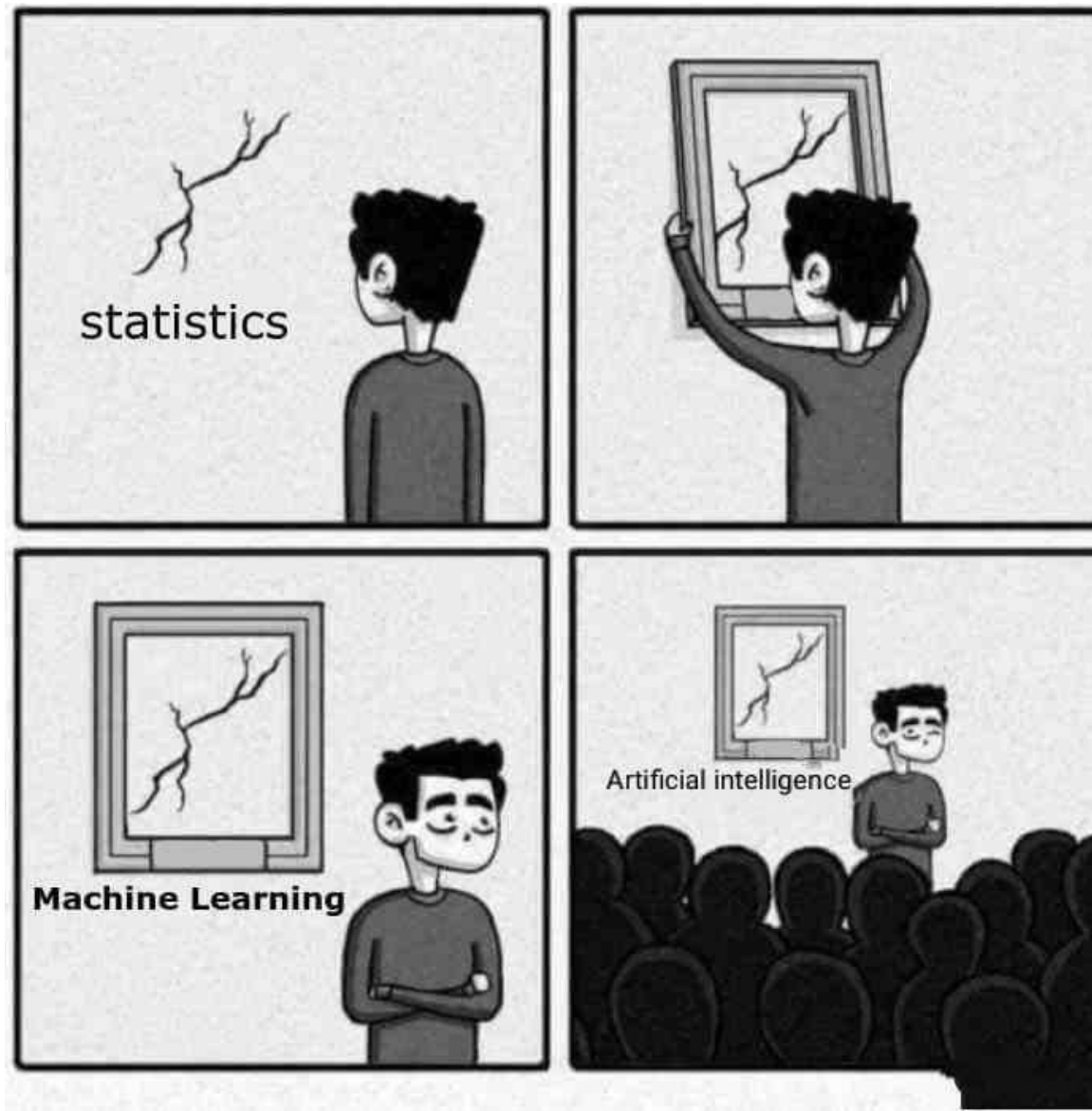
Data mining is a related field of study, focusing on exploratory data analysis through unsupervised learning.

In its application across business problems, machine learning is also referred to as predictive analytics.

Stats and ML

- ML: the same old stats, dressed up
- When you're fundraising, it's AI. When you're hiring, it's ML. When you're implementing, it's logistic regression
- Statistical learning
- ML is glorified stats
- Computer Age Statistical Inference

Stats and ML



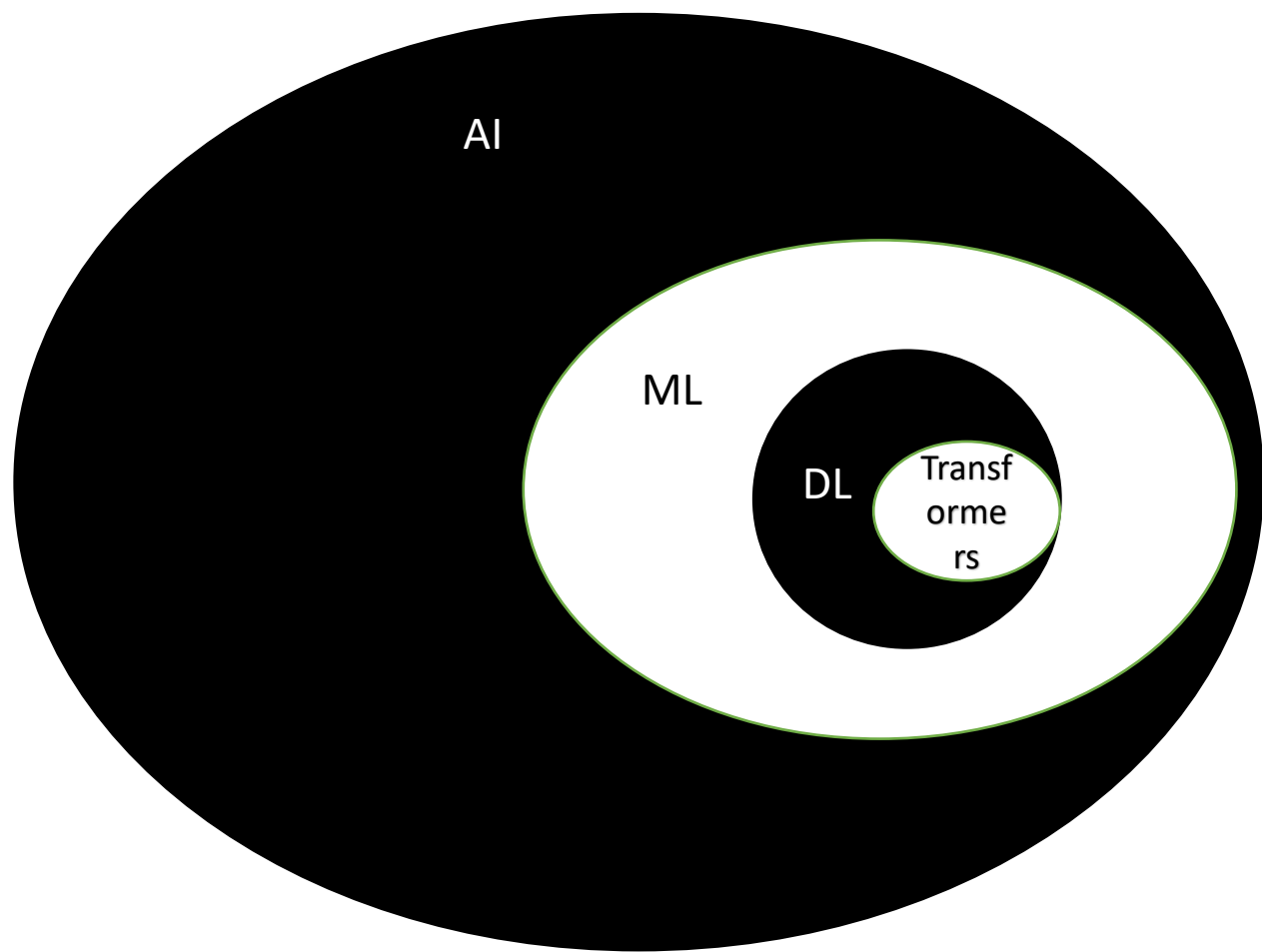
Stats vs ML. Jargon

Stats	ML
Inference	Learning
Data	Instances
Regression	Supervised learning (continuous response)
Classification	Supervised learning (discrete response)
Response	Label
Regressor	Feature
....	...

Stats vs ML. Some differences (with care)

Stat	Aprendizaje automático
Emphasis on inference	On prediction
Quantify uncertainty	Optimize
Understand underlying principles	Predict accurately
Maths	CS
More human intervention	More machine intervention

AI



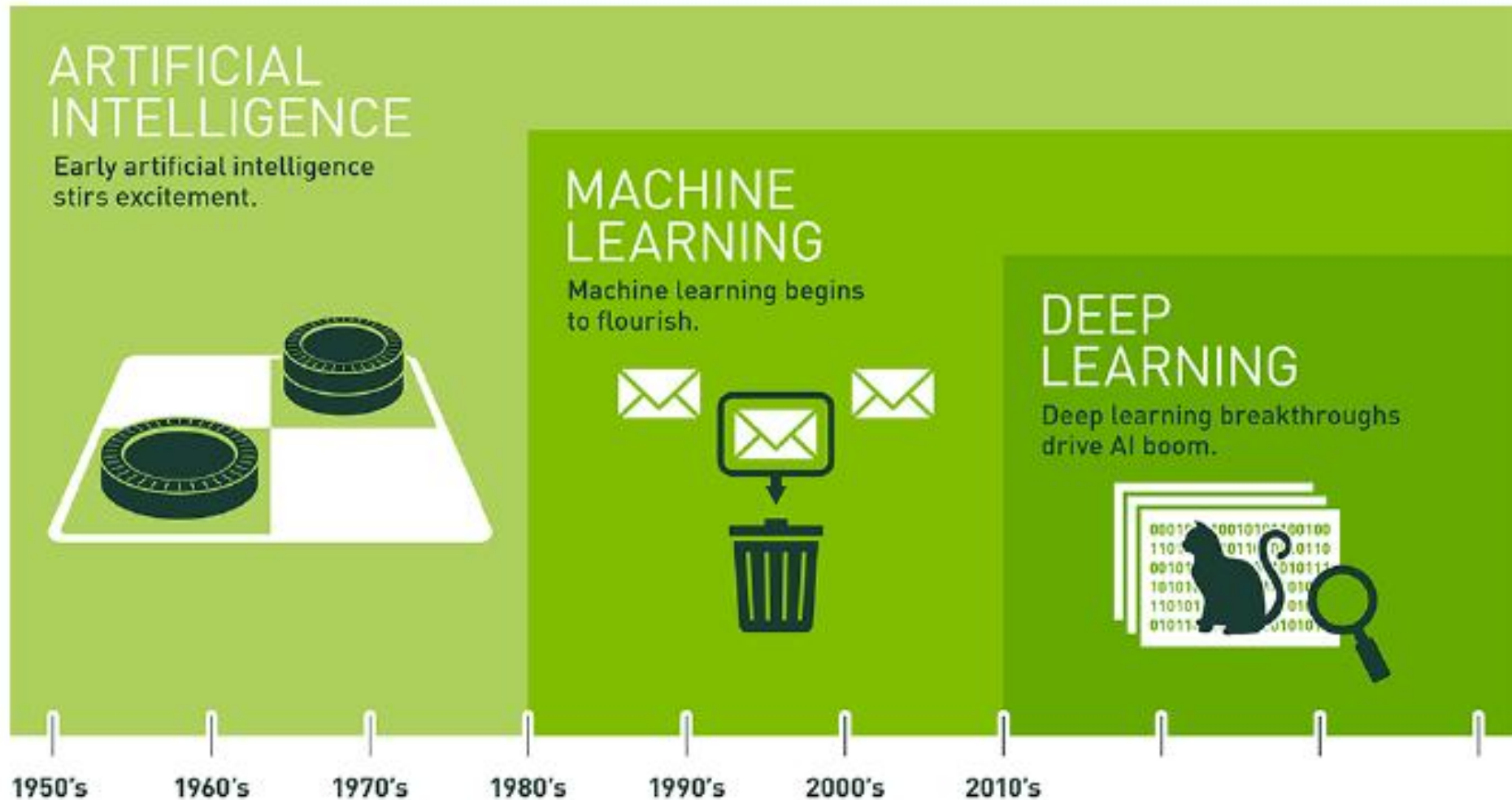
EU AI Act (Sep'23 version)

‘artificial intelligence system’ (AI system) means **software** that is developed with one or more of the techniques and approaches listed in Annex I and can, for a given set of human-defined objectives, **generate outputs such as content, predictions, recommendations, or decisions** influencing the environments they interact with;

Annex I: ML..., logic..., statistics (bayes)...

In final version: ML, logic+KB

An evolution...

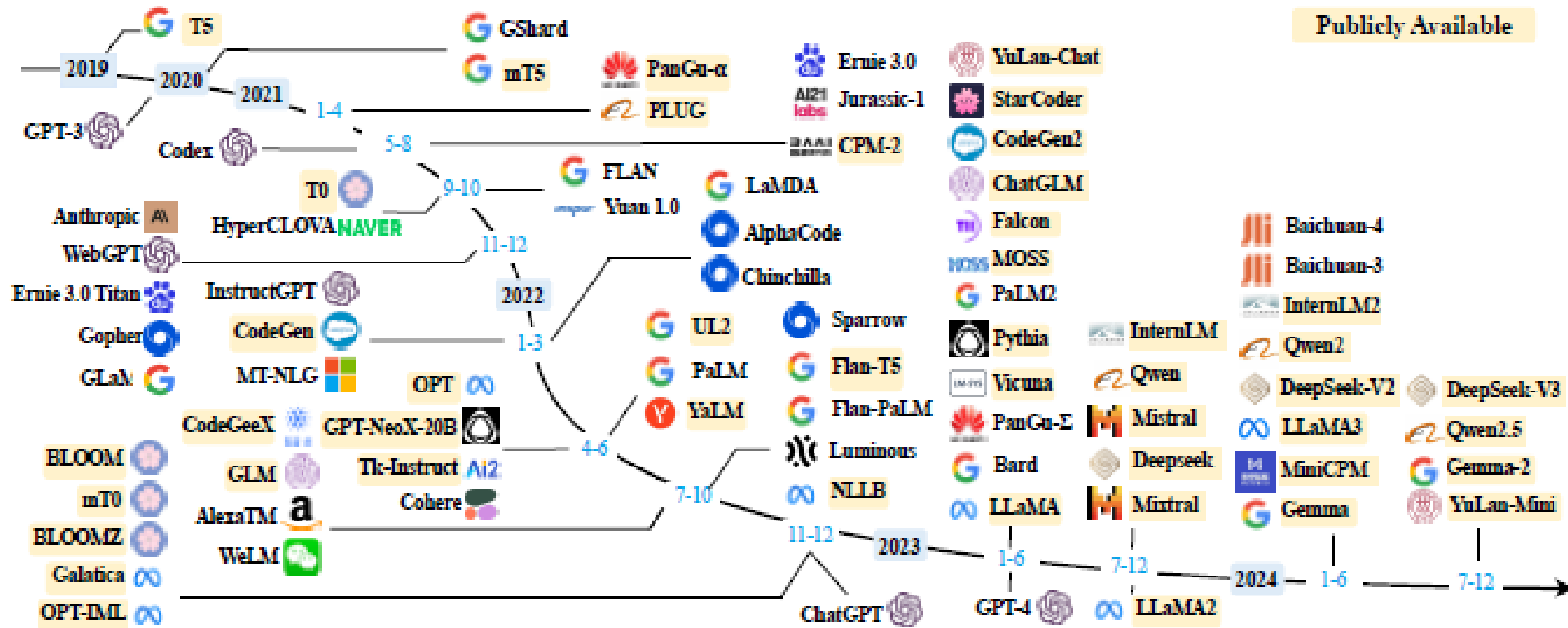


Brief history of NNs

When	What	Why	Why not
End of 50's, Beg of 60's	Rosenblatt's perceptron	Efficiente scheme Good branding	Minsky& Papert (1968)
End of 80's, Beg of 90's	Cybenko's representation Shallow NNs	Good branding Impulse from CS comm	Tech problems (vanishing gradient) Emergence of SVM and others
2010's on	Deep learning, variants Outstanding aplications	Massive labeled data Rediscovery of SGD GPUs ReLU's et al Domain specific architectures Winning Imagenet comp Transformer, LLMs	

The LLM rush

9



A survey of large language models <https://arxiv.org/abs/2303.18223>

Eight things to know about LLMs <https://arxiv.org/abs/2304.00612>

Attention is all you need (2017)

<https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html>

The LLM rainforest

- **01.AI:** Yi-34B, Yi-1.5, Yi-Large.
- **Academic:** BioBERT, FinBERT, ClinicalBERT, Legal-BERT.
- **AI21:** Jurassic-2, Jamba.
- **Alibaba:** Qwen, Qwen-7B, Qwen-VL, Qwen1.5, Qwen2, Qwen2.5, QwQ, Qwen2.5-Max, Qwen3, Qwen3.5, Tongyi Qianwen.
- **Allen Institute for AI:** SciBERT.
- **Amazon:** Amazon Titan, Amazon Nova.
- **Anthropic:** Claude 1, Claude 2, Claude Instant, Claude 3, Claude 3 Opus, Claude 3 Sonnet, Claude 3 Haiku, Claude 3.5 Sonnet, Claude 3.5 Haiku, Claude 3.7 Sonnet, Claude Opus 4, Claude Sonnet 4, Claude Opus 4.1, Claude Sonnet 4.5, Claude Haiku 4.5, Claude Opus 4.5, Claude Opus 4.6, Claude Opus 4.7, Claude Opus 4.8, Claude Sonnet 4.6, Claude Sonnet 5, Claude Fable 5.
- **Baichuan:** Baichuan, Baichuan2.
- **Baidu:** ERNIE 3.0, ERNIE Bot, ERNIE 4.0, ERNIE 4.5, ERNIE 5.0, ERNIE 5.1, Wenxin Yiyao.
- **BigCode:** StarCoder.
- **BigScience:** BLOOM language model.
- **ByteDance:** Doubao, Doubao 2.0, ByteDance Skylark.
- **Cohere:** Cohere Command, Cohere Command R, Cohere Command R+, Cohere Command A, Cohere Aya.
- **Databricks:** DBRX.
- **DeepSeek:** DeepSeek LLM, DeepSeek-Coder, DeepSeek-V2, DeepSeek-V3, DeepSeek-VL, DeepSeek-Math, DeepSeek-R1, DeepSeek-V3.1, DeepSeek-V3.2, DeepSeek-V4.
- **EleutherAI:** GPT-NeoX, GPT-J, Pythia.
- **Google:** Gopher, Chinchilla, LaMDA, Flan-T5, PaLM 2, Med-PaLM, Med-PaLM 2, Google Bard, Gemini 1.0, Gemini 1.5, Gemini 2.0, Gemini 2.5, Gemini 3, Gemini 3.5, Gemini 3.6, Gemma, Gemma 2, Gemma 3, Gemma 4, CodeGemma, BERT, ALBERT, ELECTRA, XLNet, mBERT.
- **Huawei:** Huawei PanGu.
- **Hugging Face:** DistilBERT.
- **iFlytek:** iFlytek Spark, SparkDesk.
- **Inspur:** Yuan 2.0.
- **Kunlun:** Skywork.
- **LMSYS/Academic:** Vicuna.
- **Meta:** OPT-175B, Galactica, LLaMA, Llama 2, Code Llama, Llama 3, Llama 3.1, Llama 3.2, Llama 3.3, Llama 4, Muse Spark, RoBERTa, XLM-R, XLM-RoBERTa.
- **Microsoft:** Phi-1, Phi-2, Phi-3, Phi-3.5, Phi-4, WizardLM, Orca, DeBERTa.
- **MiniMax:** MiniMax-Text, MiniMax-01, MiniMax M3.
- **Mistral:** Mistral 7B, Mixtral 8x7B, Mixtral 8x22B, Mistral Large, Mistral Large 2, Mistral Small, Mistral Medium, Mistral Nemo, Codestral, Mathstral, Ptxtral, Ministral, Mistral Small 3, Mistral Medium 3, Magistral, Mistral Large 3, Mistral Small 4, Mistral Medium 3.5, Ministral 3.
- **Moonshot AI:** Kimi Moonshot, Kimi K1.5, Kimi K2, Kimi K2.5.
- **MosaicML:** MPT-7B.
- **NVIDIA:** Nemotron, Megatron-LM.
- **OpenAI:** GPT-2, GPT-3, InstructGPT, GPT-3.5, ChatGPT, GPT-4, GPT-4 Turbo, GPT-4V, GPT-4o, GPT-4o mini, GPT-4.1, GPT-4.5, OpenAI o1, o1-mini, OpenAI o3, o3-mini, o4-mini, GPT-5, GPT-5.1, GPT-5.2, GPT-5.4, GPT-5.5, GPT-5.6.
- **SenseTime:** SenseChat, SenseNova.
- **Shanghai AI Lab:** InternLM, InternLM2, InternLM2.5.
- **Stability AI:** StableLM.
- **Stanford/Academic:** Alpaca language model.
- **StepFun:** StepFun.
- **Tencent:** Hunyuan, Hunyuan-Large, Hunyuan-T1.
- **TI:** Falcon-40B, Falcon-180B.
- **xAI:** Grok-1, Grok-1.5, Grok 2, Grok 3, Grok 4, Grok 4.5.
- **Zhipu AI:** GLM-130B, ChatGLM, ChatGLM2, ChatGLM3, GLM-4, GLM-4.5, GLM-5, GLM-5.2, CogVLM.

ML

A computer program learns from experience E with respect to task T and performance measure P ,

If its behaviour with respect to T , measured according to P , improves with experience E

Representation-Evaluation-Optimization

Goodfellow et al DL book

Some ML examples. Red matters!!!

Uncertainty is almost ubiquitous in ML:

- Given a certain transaction, is it fraudulent or not? **If fraudulent should I stop it?**
- Given the monitoring trace of an Inet device, are we facing an attack? **Should I stop operations?**
- A person with these FB likes will buy this type of beer? **Should I send him my brand add?**
- A person with these tweets is conservative? **Should I send her Brexit propaganda?**
- Robots (or ADS): If robot performs this, How will the user react? And the environment? **Consequently, what should the robot do?**

In applications, we'll need to go beyond

- Beyond a model with good fit...
- Beyond a model that predicts well...

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- Fraud detection. Classification problem
 - Few false positives. FPR
 - Few false negatives. FNR

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 - But what really matters are minimising monetary losses!!!

- Reservoir system management. Forecasting model for inputs and demands

Feeds decision model e.g to minimize energy deficit, wasted water, given constraints.....

- Aviation safety risk management. Forecasting models for accidents and incidents, as well as their multiple impacts

Feed a risk management model: optimal safety resource allocation given constraints...

- Robot control. Forecasting model for user and environments

Feeds robot control model: optimal robot decisions over time, given constraints...

ML, Stats in modern times (Big Data)

Computer Age Statistical Inference

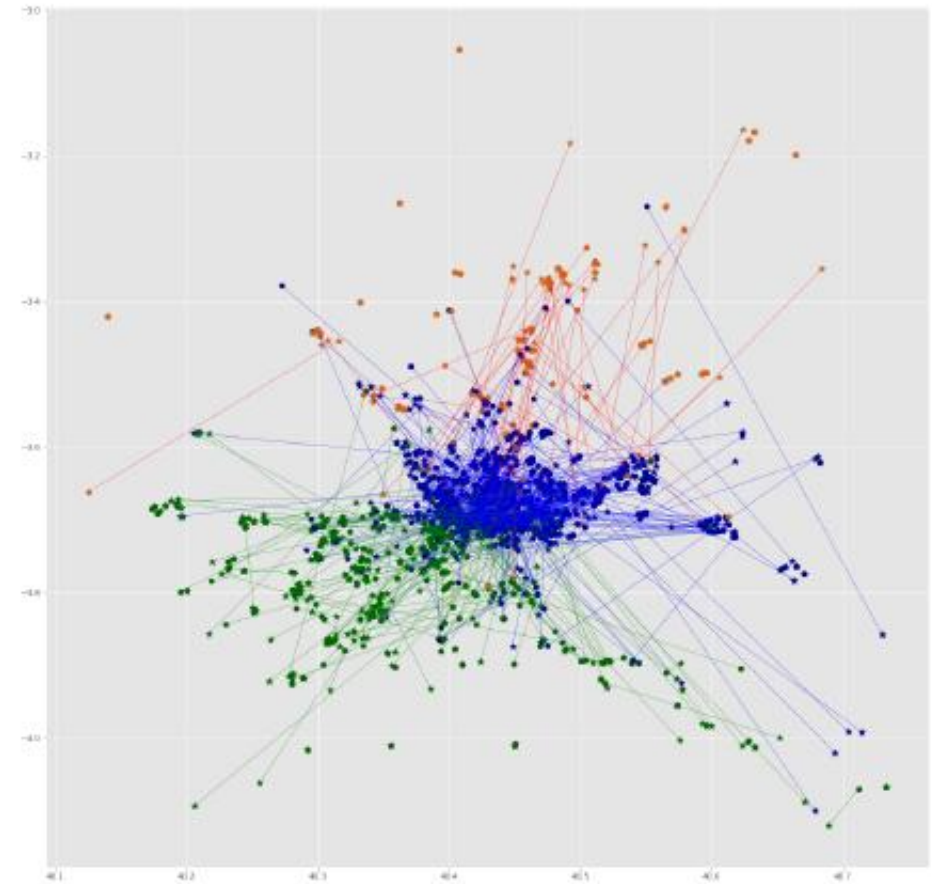
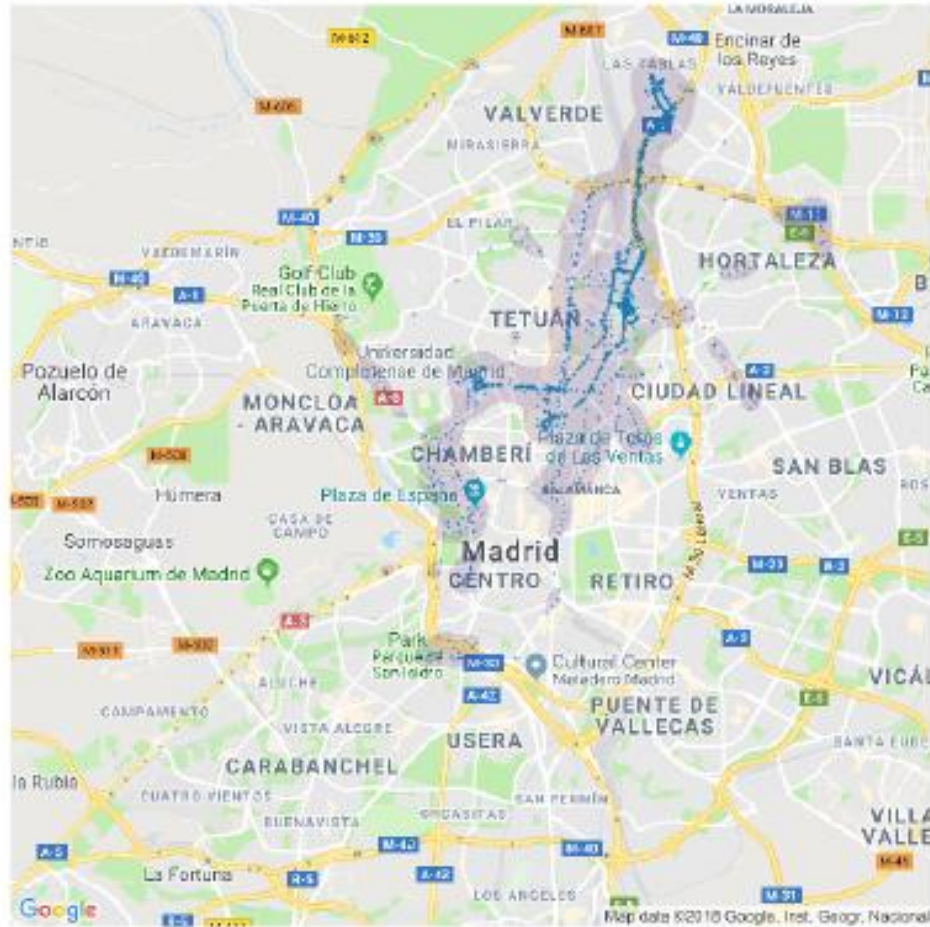
Volume. Space scalability

Variety. Text, images, sound, video,... video,....

Velocity. High frequency for data and decisions, time series, dynamic models. Time scalability

Create **v**alue by actually supporting decisions!!!

ML and BD (A marketing example)



ML: Applications, Ok but ...

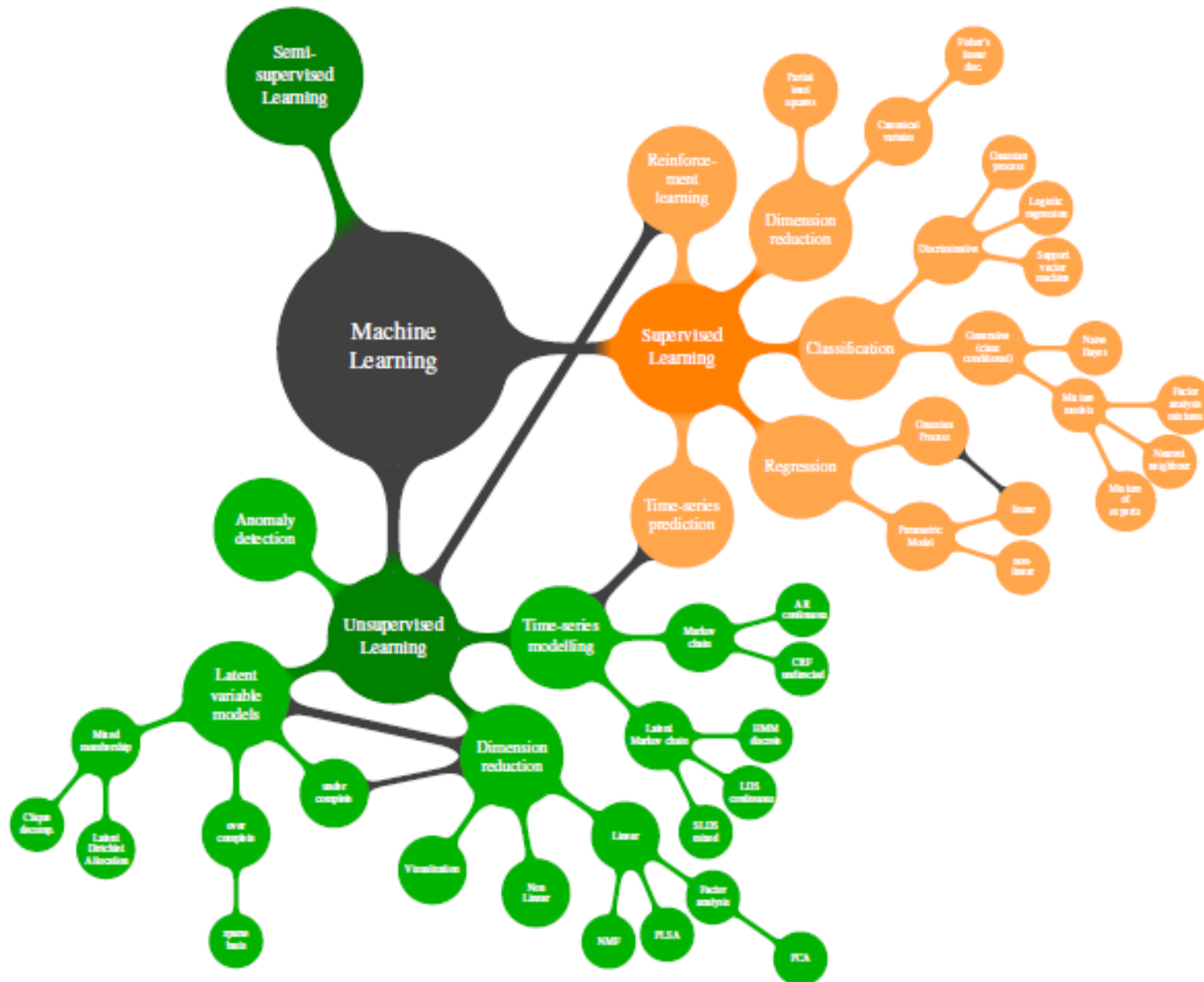
- Privacy
- Safety and security
- Fairness
- No biases
- Explainability, Interpretability
-

Core ML themes

Core ML themes

- Supervised learning: Pairs input-output available
Regression, Classification
Case: Cannabinoid detection
- Unsupervised learning: Outputs not available (or the inputs are the outputs)
Density estimation, clustering, outlier detection, Visualisation
Case: Singular community detection
- Reinforcement learning: Decisions impacting outputs on-the-fly
Markov decision processes
Case: Autonomous driving systems
- Semisupervised, self-supervised,...
- Generative vs Discriminative

With somewhat blurry borders.... (from Barber)



Into SL

SL: ingredients

- Data available: examples, samples, instances,...
- Several observed variables: predictors, attributes, features, covariates, explanatory variables, independent variables,...
- Some of special interest: response(s), dependent variable(s), target(s), output(s), label(s),...

SL: types of problems

1. Regression, response variable is continuous
2. Classification, response variable is discrete
3. Other:
 - Mixed (some continuous, some discrete)
 - Discrete but ordered
 - ...

Into UL

UL: ingredients

- Data available: examples, samples, instances,...
- Unavailable: response(s), dependent variable(s), target(s), output(s), label(s),...

Taxonomy of UL algos

- Density estimation
- Manifold learning: PCA, non-linear PCA, ...
- Finding modes and groups: cluster analysis, mixture models,...
- Sampling: GANs, Autoencoders, Variational autoencoders,...

Challenges in unsupervised learning

- High dimension of feature space
- Properties of interest more complex than parameter estimation
- No direct error quantification measure

Into RL

RL: features

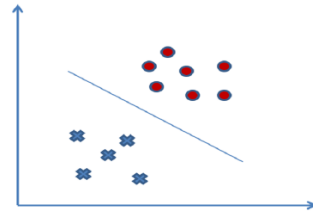
- Learning by interaction with environment
 - 'Cause-Effect' relations
 - Consequences of actions
 - What to do to achieve goals
- Goal directed learning: what to do to maximize a reward
 - Discover actions that yield most reward by trying them (trial and error search)
 - Actions affect not only immediate reward but also affect environment (delayed reward)

RL features

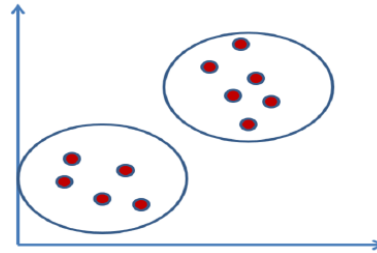
- Optimal control of incompletely known Markov decision processes
 - Schemes for sense-act-respond
 - Exploration (collect more info)-exploitation (best action)
 - Uncertainty about evolution of environment and rewards achieved
 - Sequential learning

Recap

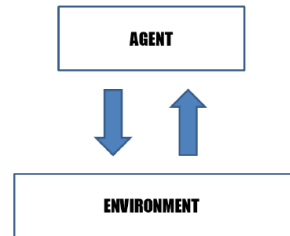
- Supervised



- Unsupervised



- Reinforced



- Semisupervised, self-supervised,....

PML and ML

PML. NI meets AI

- Bayesian inference provides a unified and coherent approach to problems of interest in Statistics, inference, prediction and **decision support**. Thus, to (most) ML problems

Yet mainstream ML focuses on MLE or MLE+regularization, check IntroML at https://datalab-icmat.github.io/courses_stats.html

- But things are changing slowly...
- PML leads to complex computational problems, some of which yet to be solved.
- An introduction to what is known (and what is yet to be discovered)
- But also intro to key Bayesian concepts
- In relation to key models in ML applications: supervised, unsupervised, reinforced, semisupervised,...

Recap: Classical vs Bayesian

Most approaches in ML (but not all, recall SVMs, RL...)

Once model fixed, we want to learn about it (its parameters)

Classical	Bayesian
Parameters fixed	Parameters uncertain, prior
Given data, formulate likelihood	Given data, formulate likelihood
Maximize likelihood to find MLE (mimimum least squares, cross entropy,...)	Aggregate likelihood and prior to get posterior
Plug in MLE to make predictions	Use predictive distribution to make predictions

Regularisers as bridges

And then used them for decision support !!!

Inference in ML

Probabilistic model of observed variables $\mathbf{x}$ and latent variables $\mathbf{z}$ (includes parameters)

$$p(\mathbf{z}, \mathbf{x})$$

ML e

$$\mathbf{z}^* = \arg \max_{\mathbf{z}} p(\mathbf{x}|\mathbf{z})$$

MAP e

$$\mathbf{z}^* = \arg \max_{\mathbf{z}} p(\mathbf{z}|\mathbf{x}) = \arg \max_{\mathbf{z}} p(\mathbf{x}|\mathbf{z})p(\mathbf{z})$$

Bayes e

$$p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{x})} = \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{\int p(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}}$$

Incorporates prior info

Estimates full distribution

Denominator (evidence) frequently intractable

MLE optimization

Problem

$$J(\theta) = \mathbb{E}_{\mathbf{x}, y \sim \hat{p}_{\text{data}}} L(\mathbf{x}, y, \theta) = \frac{1}{m} \sum_{i=1}^m L(\mathbf{x}^{(i)}, y^{(i)}, \theta)$$

$$L(\mathbf{x}, y, \theta) = -\log p(y \mid \mathbf{x}; \theta)$$

Gradient

$$\nabla_{\theta} J(\theta) = \frac{1}{m} \sum_{i=1}^m \nabla_{\theta} L(\mathbf{x}^{(i)}, y^{(i)}, \theta)$$

What if also regulariser?

What if m is large?

Stochastic gradient descent....

Beta-binomial model: A typical example

Consider recovery protocols for an SME computer service after a cyber attack. We introduce one protocol and wish to assess it, e.g. to be compared with another one.

Protocol tested in 12 attacks. Effective in 9 (e.g. attack duration was less than one hour)

Let's start with the model

A typical example

- n trials (identical, independent). Two results: success, failure
- Number X of successes in n trials
- Success probability in a trial θ_1
- Distribution of number of successes in n trials $X|\theta_1 \sim \text{Bin}(12, \theta_1)$
- For X=9,

$$\Pr(X = 9|\theta_1) \propto \theta_1^9 (1 - \theta_1)^3, \quad \theta_1 \in [0, 1]$$

A typical example

Likelihood

$$Pr(X = 9|\theta_1) \propto \theta_1^9 (1 - \theta_1)^3, \quad \theta_1 \in [0, 1]$$

First approach: Maximise likelihood --→ Maximum likelihood estimator MLE

The MLE is 9/12

But MLE has several defects...

A typical example

We may use another source of information about the parameter. The prior distribution, e.g.

$$p(\theta_1) = 1, \theta_1 \in [0, 1].$$

Update it through Bayes formula, to get the posterior

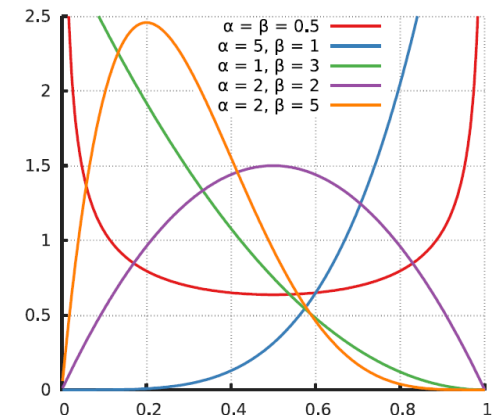
$$p(\theta_1 | x = 9) \propto p(\theta_1) \times \Pr(X = 9 | \theta_1) \propto \theta_1^9 (1 - \theta_1)^3, \theta_1 \in [0, 1]$$

which summarises all the info available about the parameter in a distribution

Beta (10,4)

Check

http://en.wikipedia.org/wiki/Beta_distribution



A typical example

The posterior serves as prior for subsequent studies. E.g., if in the following 5 applications there are 3 successes the new posterior is

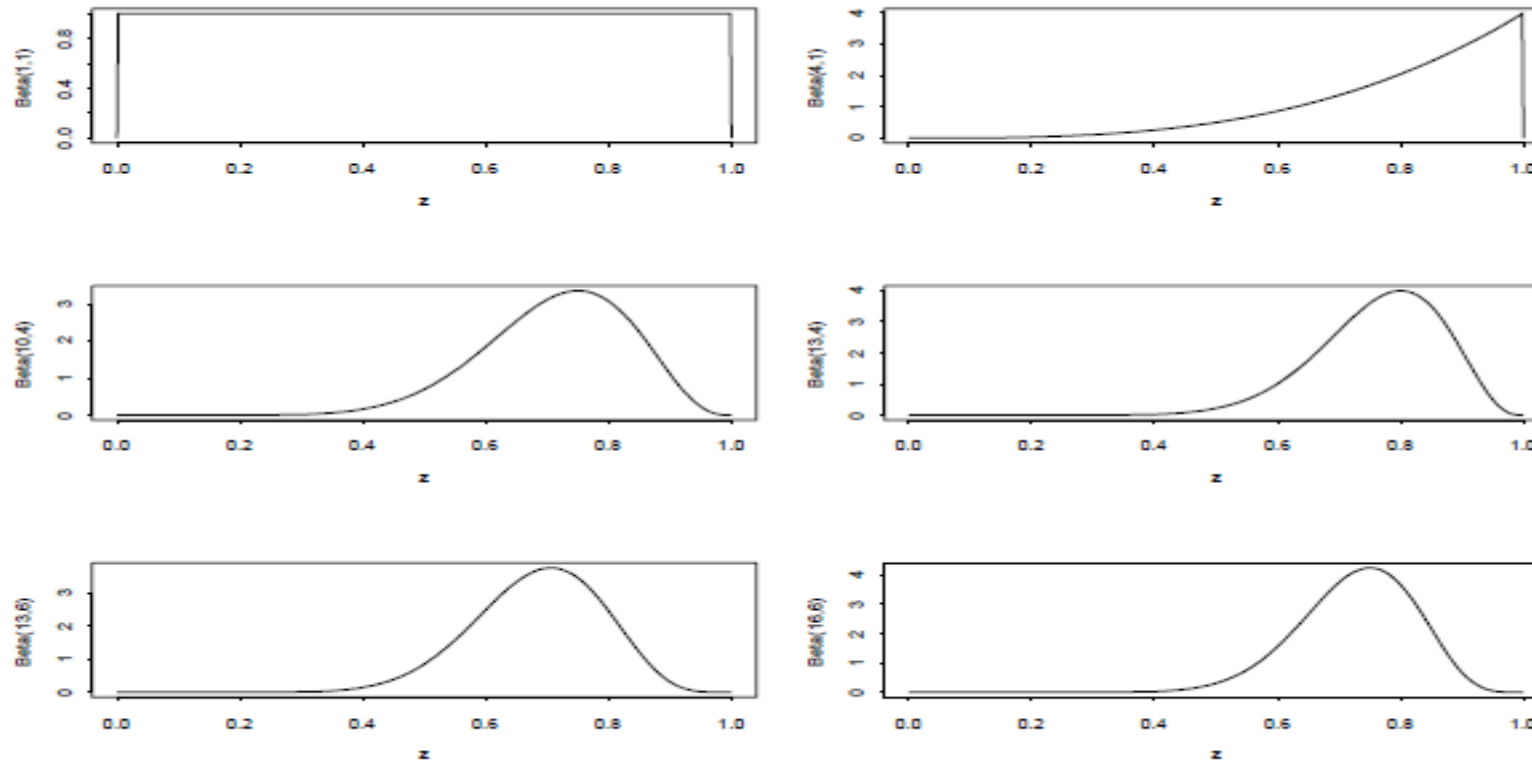
$$p(\theta_1|x = 3) \propto [\theta_1^9(1 - \theta_1)^3] \times [\theta_1^3(1 - \theta_1)^2] = \theta_1^{12}(1 - \theta_1)^5, \theta_1 \in [0, 1] \quad \text{Beta (13,6)}$$

Suppose that a priori, the probability is around 80% and bigger values are more likely, the learning goes through

$$\text{Beta (4,1)} \longrightarrow \text{Beta(13,4)} \longrightarrow \text{Beta(16,6)},$$

Sequential nature of Bayes rule

A typical example



Convergence in learning, consensus, asymptotic behavior

A typical example

- Focus on Beta (10,4). Try to use simulation for all computations also!!!

Point estimate. Summarise in a value, e.g. the posterior mean

$$\frac{10}{10+4} = 0.72$$

Why not the posterior median? Or the posterior mode (MAP)!!!

Interval estimate. Summarise interval with high probability e.g. 0.9.

- Symmetric probability wise

$$[0.505, .887]$$

- Highest posterior density interval. HDI

A typical example

- Focus on Beta (10,4)

Hypothesis testing. E.g Is the protocol effective? Null: Is the proportion bigger than 0.5
 $1 - \text{pbeta}(0.5, 10, 4) = 0.953$

Predictions Probability of more than 4 successes in 7 trials

$$\begin{aligned} Pr(X = k | x = 9) &= \int Pr(X = k | \theta_1) p(\theta_1 | x = 9) d\theta_1 = \\ &= \int \binom{7}{k} \theta_1^k (1 - \theta_1)^{7-k} \binom{13}{3} \theta_1^9 (1 - \theta_1)^3 d\theta_1 = \\ &= \frac{\binom{7}{k} \binom{13}{3}}{\binom{20}{9+k}}. \end{aligned}$$

$$Pr(X \geq 5 | x = 9) = \sum_{k=5}^7 Pr(X = k | x = 9) = 0.6641.$$

A typical example

Consider a second protocol. 10 opportunities, successful in 6. θ_2

Model

$$X|\theta_1 \sim \text{Bin}(12, \theta_1)$$

$$Y|\theta_2 \sim \text{Bin}(10, \theta_2)$$

$$\theta_1, \theta_2 \sim \text{Unif}[0, 1]$$

independent

Want

$$r = \text{Pr}(\theta_1 \geq \theta_2 | x = 9, y = 6)$$

A typical example

$$\theta_1 \sim \text{Beta}(10, 4), \theta_2 \sim \text{Beta}(7, 5)$$

- Distribution of $\theta_1 - \theta_2$????
- Through simulation. E.g 1000 observations, compute differences, count those bigger than 0, divide by 1000.
- Which protocol is better?

$$r \approx 0.772.$$

A typical example

- Utility structure

	succeeds	does not succeed
Plan A	0.8	0
Plan B	1	0.2

- Expected utilities given probabilities

$$0.8\theta_1 + 0(1 - \theta_1) = 0.8\theta_1$$

$$\theta_2 + 0.2(1 - \theta_2) = 0.2 + 0.8\theta_2$$

- Expected utilities

$$0.8E(\theta_1|x = 9) = 0.8 \times \frac{10}{14} = \frac{4}{7} \qquad 0.2 + 0.8E(\theta_2|y = 6) = 0.2 + 0.8 \times \frac{7}{12} = \frac{2}{3}.$$

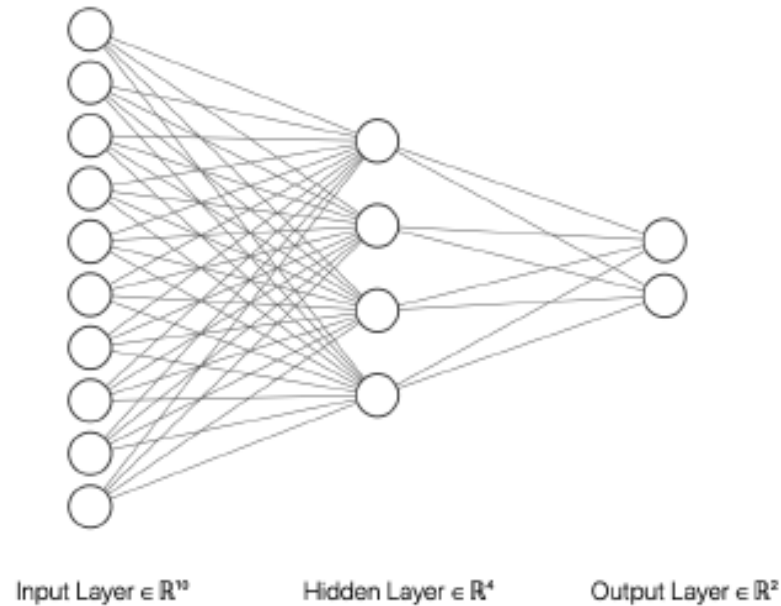
Recap. Computational problems in BML

Plagued by complex integrals with complex integrands
+ optimisations

Easy conceptually... tough computationally

PML and ML through NNs

Shallow NNs. Formulation



$$y = \sum_{j=1}^m \beta_j \psi(x' \gamma_j) + \epsilon$$

$$\epsilon \sim N(0, \sigma^2),$$

$$\psi(\eta) = \exp(\eta) / (1 + \exp(\eta))$$

Linear in β 's, nonlinear in γ 's

Training

Given training data, maximise log-likelihood

$$\min_{\beta, \gamma} f(\beta, \gamma) = \sum_{i=1}^n f_i(\beta, \gamma) = \sum_{i=1}^n \left(y_i - \sum_{j=1}^m \beta_j \psi(x'_i \gamma_j) \right)^2$$

Gradient descent

Backpropagation to estimate gradient

Training with regularisation

$$\min_{\beta, \gamma} f(\beta, \gamma) = \sum_{i=1}^n f_i(\beta, \gamma) = \sum_{i=1}^n \left(y_i - \sum_{j=1}^m \beta_j \psi(x'_i \gamma_j) \right)^2$$

$$\min g(\beta, \gamma) = f(\beta, \gamma) + h(\beta, \gamma),$$

Weight decay

$$h(\beta, \gamma) = \lambda_1 \sum \beta_i^2 + \lambda_2 \sum \sum \gamma_{ji}^2$$

Ridge

Gradient descent

$$(\beta, \gamma)_{k+1} = (\beta, \gamma)_k - \eta \nabla g((\beta, \gamma)_k)$$

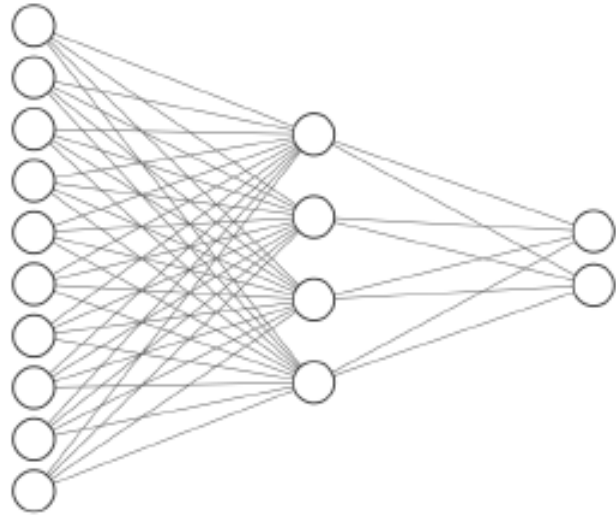
$$\nabla g((\beta, \gamma)) = \sum_{i=1}^n \nabla f_i(\beta, \gamma) + \nabla h(\beta, \gamma)$$

$$(\nabla_{\beta} f_i)_k = -2 \left(y_i - \sum_{j=1}^m \beta_j \psi(x'_i \gamma_j) \right) \psi(x'_i \gamma_k)$$

$$(\nabla_{\gamma} f_i)_{k,l} = -2 \left(y_i - \sum_{j=1}^m \beta_j \psi(x'_i \gamma_j) \right) \beta_l \psi(x'_i \gamma_l) (1 - \psi(x'_i \gamma_l)) x_k$$

$$(\nabla_{\beta} h)_k = 2\lambda_1 \beta_k \quad (\nabla_{\gamma} h)_{k,l} = 2\lambda_2 \gamma_{k,l}.$$

Deep NNs. Concept



Input Layer $\in \mathbb{R}^8$

Hidden Layer $\in \mathbb{R}^4$

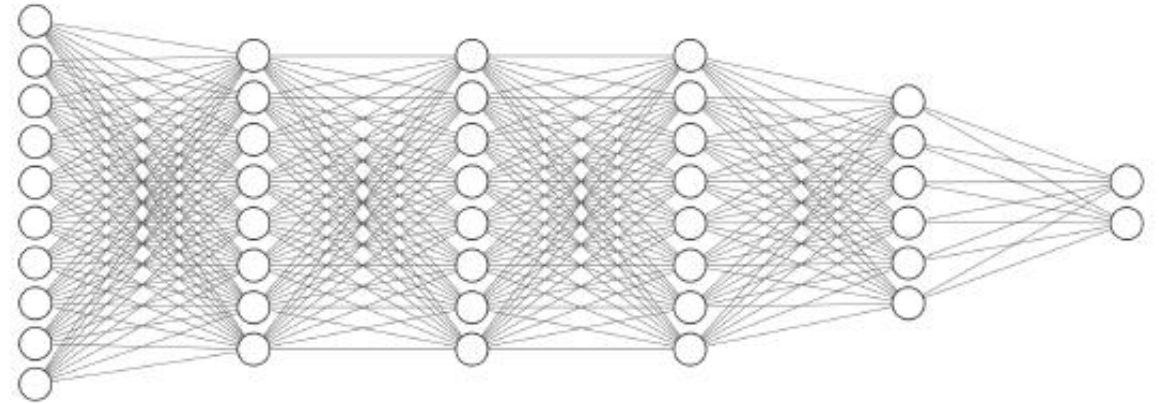
Output Layer $\in \mathbb{R}^2$

$$y = \sum_{j=1}^m \beta_j \psi(x' \gamma_j) + \epsilon$$

$$\epsilon \sim N(0, \sigma^2),$$

$$\psi(\eta) = \exp(\eta) / (1 + \exp(\eta))$$

(Shallow) Neural nets



$$\{f_0, f_1, \dots, f_{L-1}\}$$

$$z_{l+1} = f_l(z_l, \gamma_l).$$

$$y = \sum_{j=1}^{m_L} \beta_j z_{L,j} + \epsilon$$

$$\epsilon \sim N(0, \sigma^2),$$

Deep neural nets

Backpropagation (CASI 18)

Algorithm 18.1 BACKPROPAGATION

- 1 Given a pair x, y , perform a “feedforward pass,” computing the activations $a_\ell^{(k)}$ at each of the layers $L_2, L_3, \dots, L_K$; i.e. compute $f(x; \mathcal{W})$ at x using the current $\mathcal{W}$, saving each of the intermediary quantities along the way.

- 2 For each output unit ℓ in layer L_K , compute

$$\begin{aligned}\delta_\ell^{(K)} &= \frac{\partial L[y, f(x, \mathcal{W})]}{\partial z_\ell^{(K)}} \\ &= \frac{\partial L[y, f(x; \mathcal{W})]}{\partial a_\ell^{(K)}} \dot{g}^{(K)}(z_\ell^{(K)}),\end{aligned}\tag{18.10}$$

where $\dot{g}$ denotes the derivative of $g(z)$ wrt z . For example for $L(y, f) = \frac{1}{2} \|y - f\|_2^2$, (18.10) becomes $-(y_\ell - f_\ell) \cdot \dot{g}^{(K)}(z_\ell^{(K)})$.

- 3 For layers $k = K - 1, K - 2, \dots, 2$, and for each node ℓ in layer k , set

$$\delta_\ell^{(k)} = \left(\sum_{j=1}^{p_{k+1}} w_{j\ell}^{(k)} \delta_j^{(k+1)} \right) \dot{g}^{(k)}(z_\ell^{(k)}).\tag{18.11}$$

- 4 The partial derivatives are given by

$$\frac{\partial L[y, f(x; \mathcal{W})]}{\partial w_{\ell j}^{(k)}} = a_j^{(k)} \delta_\ell^{(k+1)}.\tag{18.12}$$

From gradient descent...

Training goes through minimising

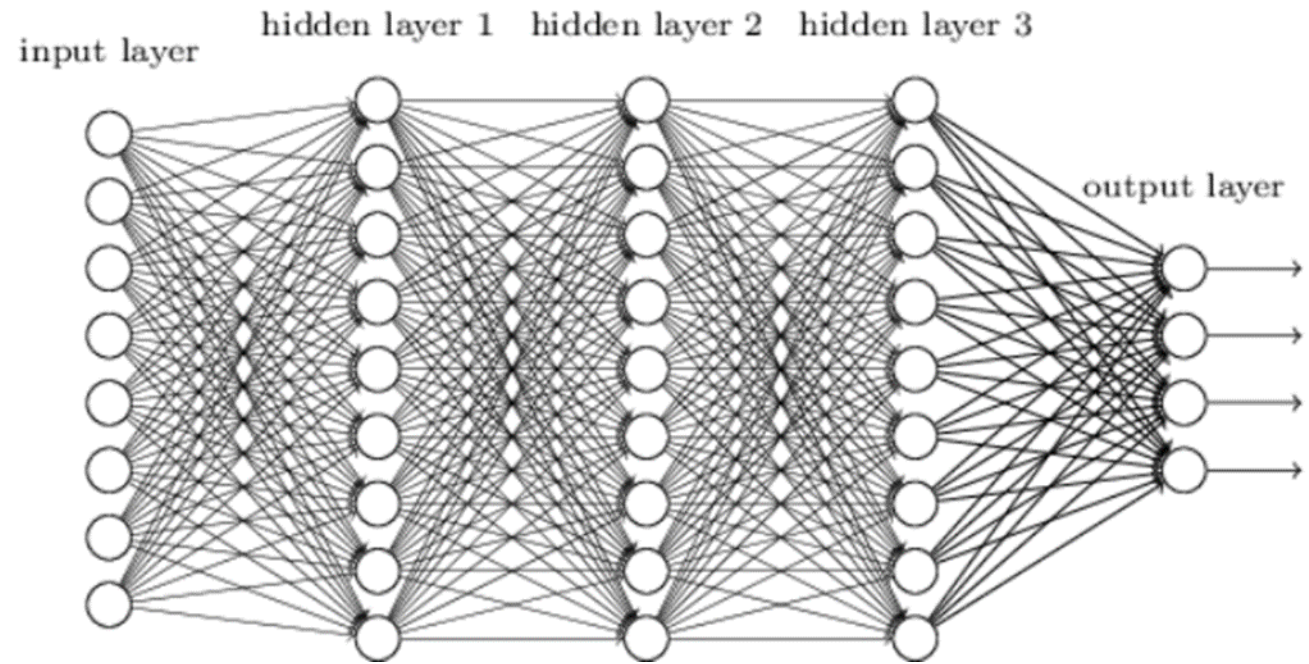
$$J(\theta) = \mathbb{E}_{\mathbf{x}, y \sim \hat{p}_{\text{data}}} L(\mathbf{x}, y, \theta) = \frac{1}{m} \sum_{i=1}^m L(\mathbf{x}^{(i)}, y^{(i)}, \theta)$$

$$L(\mathbf{x}, y, \theta) = -\log p(y \mid \mathbf{x}; \theta)$$

Requires gradient

$$\nabla_{\theta} J(\theta) = \frac{1}{m} \sum_{i=1}^m \nabla_{\theta} L(\mathbf{x}^{(i)}, y^{(i)}, \theta)$$

Might not even fit in memory, very slow anyway



... to stochastic gradient descent

(Randomly) sample a minibatch of size m' .

Approximate gradient

$$\mathbf{g} = \frac{1}{m'} \nabla_{\boldsymbol{\theta}} \sum_{i=1}^{m'} L(\mathbf{x}^{(i)}, y^{(i)}, \boldsymbol{\theta})$$

Update via gradient descent

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \epsilon \mathbf{g}$$

What is to be gained?

- Uncertainties in predictions
- Improved decision making based on above (risk aversion etc...)
- Some explainability via hypothesis testing
- Architecture choice
- Incorporating prior info (at least structurally)

Bayesian analysis of shallow neural nets (fixed arch)

$$y = \sum_{j=1}^m \beta_j \psi(\mathbf{x}' \gamma_j) + \epsilon$$

$$\epsilon \sim N(0, \sigma^2),$$

$$\psi(\eta) = \exp(\eta) / (1 + \exp(\eta))$$

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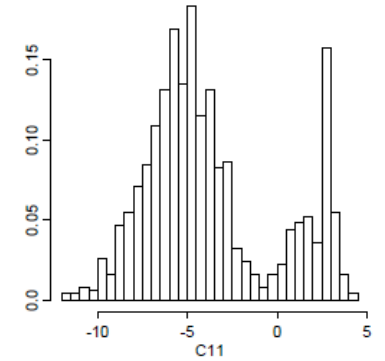
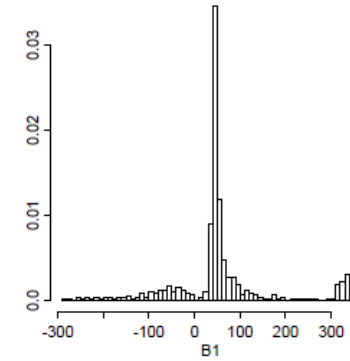
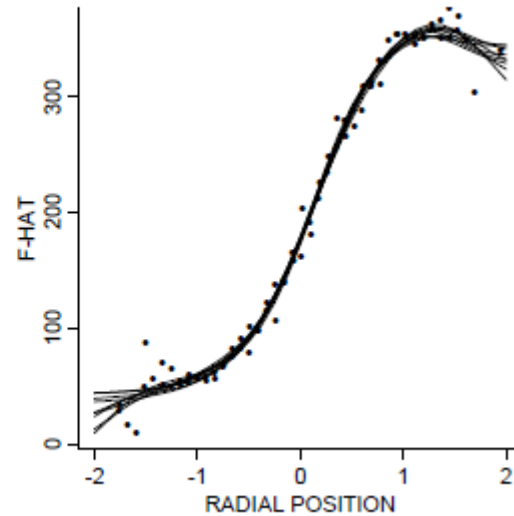
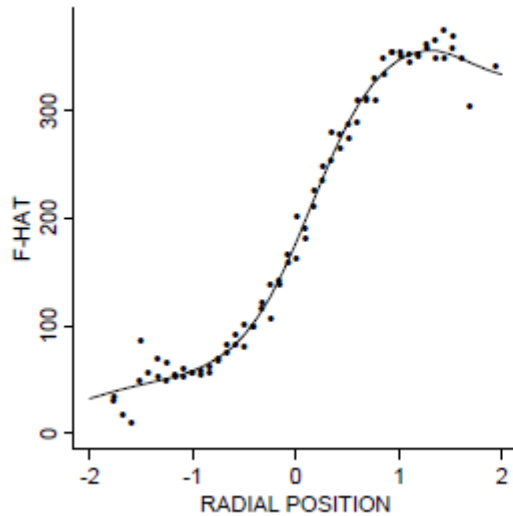
$$\beta_i \sim N(\mu_\beta, \sigma_\beta^2) \text{ and } \gamma_i \sim N(\mu_\gamma, S_\gamma^2)$$

$$\mu_\beta \sim N(a_\beta, A_\beta), \mu_\gamma \sim N(a_\gamma, A_\gamma), \sigma_\beta^{-2} \sim \text{Gamma}(c_b/2, c_b C_b/2)$$

$$S_\gamma^{-1} \sim \text{Wish}(c_\gamma, (c_\gamma C_\gamma)^{-1}) \text{ and } \sigma^{-2} \sim \text{Gamma}(s/2, sS/2)$$

Uncertainty in predictions, explainability

$$\hat{f}(x) = \hat{E}(y_{n+1}|x_{n+1}, D) = \frac{1}{k} \sum_{t=1}^k E(y_{N+1}|x_{n+1}, \theta = \theta_t)$$



Bayesian analysis of shallow neural nets (var arch)

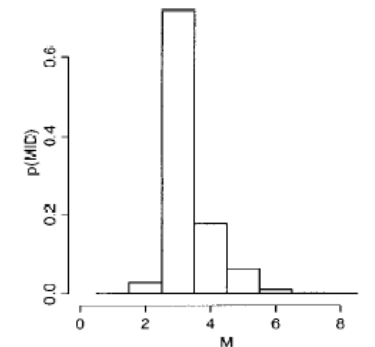
$$y = x'_i a + \sum_{j=1}^{m^*} d_j \beta_j \psi(x' \gamma_j) + \epsilon$$

$$\epsilon \sim N(0, \sigma^2),$$

$$\psi(\eta) = \exp(\eta) / (1 + \exp(\eta)),$$

$$Pr(d_j = k | d_{j-1} = 1) = (1 - \alpha)^{1-k} \times \alpha^k, k \in \{0, 1\}$$

$$\beta_i \sim N(\mu_b, \sigma_\beta^2), a \sim N(\mu_a, \sigma_a^2), \gamma_i \sim N(\mu_\gamma, \Sigma_\gamma).$$



Reversible jump algo, Bayesian model averaging

The road ahead

Contents

1. Intro. Overview ML and PML
2. Generative models. From beta-binomial and normal-normal to modern generative models
3. Bayesian regression (including logistic) and DLMs
4. Computational Bayesian methods
5. Bayesian neural nets
6. A case study (with a focus on PGMs)

All illustrated with R and rstan

ML and Stats. Easy reading

- J. Friedman. Data Mining and Statistics, What's the connection (1998)
- L. Breiman. Statistical Modeling,. The two cultures (2001)
- Cross Validated. What's the difference between data mining, statistics, machine learning and AI (2010)
- S.D. Sekar What's the difference between Artificial Intelligence, Machine Learning, Statistics and Big Data (2014)
- Cross Validated. What exactly is Big Data? (2015)
- David Donoho. 50 years of data science (2015)
- B. Efron, T. Hastie. Computer Age Statistical Inference (2016)
- *D. Dunson Statistics in the Big Data era: Failures of the Machine* (2019)
- D. Spiegelhalter The Art of Statistics (2020)
- *M. Hernan, J. Hsu, B. Healy A second chance to get causal inference right: a classification of data science tasks* (2019)

ML

- Efron, Hastie (2017) Computer Age Statistical Inference. Camb. UP
- Goodfellow, Bengio, Courville (2017) Deep Learning, MIT Press.
- Hastie, Tibshirani, Friedman (2009) Elements of Statistical Learning. Springer
- James, Hastie, Witten, Tibshirani (2013) An intro to Statistical Learning. Springer.
- James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). An intro to Statistical Learning: with Applications in Python. Springer

https://datalab-icmat.github.io/courses_stats.html#Introduction to Machine Learning
<https://lms-cunef-icmat-rg2024.github.io/>

Bayes

- French, DRI (2000) Statistical Decision Theory, Wiley
- Hoff, P. (2009) A first course in Bayesian Statistical Methods, Springer
- Berger, J. (2013) Statistical decision theory and Bayesian analysis, Springer
- Robert, C. (2007) The Bayesian choice, Springer
- Gelman, A., Carlin, J..... (2013) Bayesian Data Analysis, CRC
- DRI,Ruggeri, Wiper (2012) BASP, Wiley
- Agresti, Kateri, Grove, Mira (2026) Bayesian Stats: A primer for data scientists with Python and R

[https://datalab-icmat.github.io/courses_stats.html#Bayesian Data Science](https://datalab-icmat.github.io/courses_stats.html#Bayesian_Data_Science)

PML/Bayes and ML

- Barber (2020) Bayesian reasoning and machine learning.
- Bishop (2006) Pattern Recognition and Machine Learning. Springer
- Murphy (2014, 2022, 2023) PML
- Naveiro, DRI (2026) BAML

<https://pml-reading-club.github.io/resources.html>

ML and R

- Chollet (2018) Deep Learning with R, Manning.
- Ciaburro (2020) Reinforcement Learning with R, Packt
- Rstan manuals
- Many of the earlier

Lab 1. Remember R